

Nicklas Hansen

hello@nicklashansen.com | nicklashansen.com | [nicklashansen](https://github.com/nicklashansen) | [ncklas](https://www.linkedin.com/in/ncklas) | [San Diego, CA](#)

Education

University of California, San Diego

PhD student, Computer Science and Engineering, GPA: 3.85/4.0
 · Advised by Xiaolong Wang and Hao Su. NVIDIA Graduate Fellow.

San Diego, CA, USA

Sep 2021 - present

University of California, Berkeley

Visiting Student, GPA: 4.0/4.0
 · Spar Nord Fonden's FinTech scholarship recipient. SCET's Collider Cup finalist.

Berkeley, CA, USA

Spring 2020

Technical University of Denmark

MS Mathematical Modeling & Computation, GPA: 11.2/12.0
 · Special topics in machine learning. Advised by Ole Winther.

Kongens Lyngby, Denmark

Feb 2019 - Feb 2021

Technical University of Denmark

BS Software Technology, GPA: 8.2/12.0, final year GPA: 10.8/12.0
 · **Nanyang Technological University, Singapore** - semester abroad, Fall 2017.

Kongens Lyngby, Denmark

Sep 2015 - Dec 2018

Publications & Preprints (35)

Hallucination in World Models is Predictable and Preventable

Paper under review

Nicklas Hansen, Xiaolong Wang

<https://www.nicklashansen.com/mmbench2/paper.pdf>

Under review

2026

HiGym: Learning to Act in High-Dimensional Spaces

Paper under review

Harshit Dhankhar, Xiaolong Wang, **Nicklas Hansen**

preprint available soon

Under review

2026

Generating Robot Hands from Human Demonstrations

Paper under review

Yi Sha, **Nicklas Hansen**, Xueqian Bai, Carmelo Sferrazza, Michael T. Tolley, Xiaolong Wang

preprint available soon

Under review

2026

Catch It All!: Generalizable Dynamic Catching with a Dexterous Hand

Paper under review

Arth Shukla*, Kehlani Fay*, Huaze Liu[^], Darin Djapri[^], Alexiy Buynitsky, **Nicklas Hansen**

preprint available soon

Under review

2026

RoboTALES: Learning Reasoning-Guided Robot Policies via Task-Aligned Simulated Futures

Paper under review

Hanan Gari, Tejal Kulkarni, Madhoolika Venkata Chodavarapu, **Nicklas Hansen**, Manmohan Chandraker

preprint available soon

Under review

2026

Abstracting Task-Centric World Models from Visual Foundations

Paper under review

Minghao Fu, Fan Feng, **Nicklas Hansen**, Biwei Huang

preprint available soon

Under review

2026

Bayesian Curriculum Learning over Latent Geometry in Large Language Models

Paper under review

Darrien McKenzie, **Nicklas Hansen**, Xiaolong Wang
preprint available soon

Under review

2026

Leveraging Physics-Based Models for Rapid Adaptation in Reinforcement Learning

Paper under review

Adrian Remonda, Jiajun Xi, **Nicklas Hansen**, Marcus Greiff, John Talbot, John Subosits, Xiaolong Wang
preprint available soon

Under review

2026

TIPS: Turn-Level Information-Potential Reward Shaping for Search-Augmented LLMs

International Conference on Learning Representations (ICLR)

Yutao Xie, Nathaniel Thomas, **Nicklas Hansen**, Yang Fu, Li Erran Li, Xiaolong Wang
<https://arxiv.org/abs/2603.22293>

2026

Learning Massively Multitask World Models for Continuous Control

International Conference on Learning Representations (ICLR)

Nicklas Hansen, Hao Su*, Xiaolong Wang*
<https://arxiv.org/abs/2511.19584>

2026

Learning to Design Soft Hands using Reward Models

International Conference on Robotics and Automation (ICRA)

Xueqian Bai, **Nicklas Hansen**, Adabhav Singh, Michael T. Tolley, Yan Duan, Pieter Abbeel, Xiaolong Wang, Sha Yi
<https://arxiv.org/abs/2510.17086>

2025

MyoChallenge 2024: A New Benchmark for Physiological Dexterity and Agility in Bionic Humans

Conference on Neural Information Processing Systems (NeurIPS)

Cheryl Wang, Chun Kwang Tan, [...], **Nicklas Hansen**, [...], Vikash Kumar, Vittorio Caggiano (26 authors)
<https://openreview.net/forum?id=1dSLbhErNv>

2025

Generalizable Robotic Insertion with World Models

Workshop on Out-of-Distribution Generalization in Robotics @ RSS

Nicklas Hansen, Iretiayo Akinola, Yijie Guo, Jie Xu, Bingjie Tang, Hao Su, Xiaolong Wang, Abhishek Gupta, Dieter Fox, Yashraj Narang
<https://openreview.net/pdf?id=DR3n6IqGKI>

2025

Multi-Stage Manipulation with Demonstration-Augmented Reward, Policy, and World Model Learning

International Conference on Machine Learning (ICML)

Adrià López Escoriza, **Nicklas Hansen**, Stone Tao, Tongzhou Mu, Hao Su
<https://arxiv.org/abs/2503.01837>

2025

Policy Learning with Multi-Task World Models

International Conference on Learning Representations (ICLR)

Ignat Georgiev, Varun Giridhar, **Nicklas Hansen**, Animesh Garg
<https://arxiv.org/abs/2407.02466>

2025

Hierarchical World Models as Visual Whole-Body Humanoid Controllers

International Conference on Learning Representations (ICLR)

Nicklas Hansen, Jyothir S V, Vlad Sobal, Yann LeCun, Xiaolong Wang*, Hao Su*
<https://arxiv.org/abs/2405.18418>

2025

A Simulation Benchmark for Autonomous Racing with Large-Scale Human Data

Conference on Neural Information Processing Systems (NeurIPS)

Adrian Remonda, **Nicklas Hansen**, Ayoub Raji, Nicola Musiu, Marko Bertogna, Eduardo E. Veas, Xiaolong Wang
<https://arxiv.org/abs/2407.16680>

2024

A Recipe for Unbounded Data Augmentation in Visual Reinforcement Learning

Reinforcement Learning Conference (RLC)

Abdulaziz Almuzairee, **Nicklas Hansen**, Henrik I. Christensen

<https://arxiv.org/abs/2405.17416>

2024

TD-MPC2: Scalable, Robust World Models for Continuous Control

International Conference on Learning Representations (ICLR)

Nicklas Hansen, Hao Su*, Xiaolong Wang*

<https://arxiv.org/abs/2310.16828>

Spotlight

2024

Open X-Embodiment: Robotic Learning Datasets and RT-X Models

International Conference on Robotics and Automation (ICRA)

Open X-Embodiment Collaboration, [...], **Nicklas Hansen**, [...] (173 authors)

<https://arxiv.org/abs/2310.08864>

Best Conference Paper

2024

MoDem-V2: Visuo-Motor World Models for Real-World Robot Learning

International Conference on Robotics and Automation (ICRA)

Patrick Lancaster, **Nicklas Hansen**, Aravind Rajeswaran, Vikash Kumar

<https://arxiv.org/abs/2309.14236>

2024

Finetuning Offline World Models in the Real World

Conference on Robot Learning (CoRL)

Yunhai Feng*, **Nicklas Hansen***, Ziyang Xiong*, Chandramouli Rajagopalan, Xiaolong Wang

<https://arxiv.org/abs/2310.16029>

Oral

2023

Multi-Task Real Robot Learning with Generalizable Neural Feature Fields

Conference on Robot Learning (CoRL)

Yanjie Ze, Ge Yan, Yueh-Hua Wu, Annabella Macaluso, Yuying Ge, Jianglong Ye, **Nicklas Hansen**, Li Erran Li, Xiaolong Wang

<https://arxiv.org/abs/2308.16891>

Oral

2023

On Pre-Training for Visuo-Motor Control: Revisiting a Learning-from-Scratch Baseline

International Conference on Machine Learning (ICML)

Nicklas Hansen*, Zhechen Yuan*, Yanjie Ze*, Tongzhou Mu*, Aravind Rajeswaran^, Hao Su^, Huazhe Xu^, Xiaolong Wang^

<https://arxiv.org/abs/2212.05749>

2023

MoDem: Accelerating Visual Model-Based Reinforcement Learning with Demonstrations

International Conference on Learning Representations (ICLR)

Nicklas Hansen, Yixin Lin, Hao Su, Xiaolong Wang, Vikash Kumar, Aravind Rajeswaran

<https://arxiv.org/abs/2212.05698>

2023

On the Feasibility of Cross-Task Transfer with Model-Based Reinforcement Learning

International Conference on Learning Representations (ICLR)

Yifan Xu*, **Nicklas Hansen***, Zirui Wang, Yung-Chieh Chan, Hao Su, Zhouwen Tu

<https://arxiv.org/abs/2210.10763>

2023

Visual Reinforcement Learning with Self-Supervised 3D Representations

IEEE Robotics and Automation Letters (RA-L)

International Conference on Intelligent Robots and Systems (IROS)

Yanjie Ze*, **Nicklas Hansen***, Yinbo Chen, Mohit Jain, Xiaolong Wang

<https://arxiv.org/abs/2210.07241>

2023

Graph Inverse Reinforcement Learning from Diverse Videos

Conference on Robot Learning (CoRL)

Sateesh Kumar, Jonathan Zamora*, **Nicklas Hansen***, Rishabh Jangir, Xiaolong Wang

<https://arxiv.org/abs/2207.14299>

Oral

2022

Temporal Difference Learning for Model Predictive Control

International Conference on Machine Learning (ICML)

Nicklas Hansen, Xiaolong Wang*, Hao Su*

<https://arxiv.org/abs/2203.04955>

Spotlight

2022

Look Closer: Bridging Egocentric and Third-Person Views with Transformers for Robotic Manipulation

IEEE Robotics and Automation Letters (RA-L)

International Conference on Robotics and Automation (ICRA)

Rishabh Jangir*, Nicklas Hansen*, Sambaran Ghosal, Mohit Jain, Xiaolong Wang

<https://arxiv.org/abs/2201.07779>

2022

Learning Vision-Guided Quadrupedal Locomotion End-to-End with Cross-Modal Transformers

International Conference on Learning Representations (ICLR)

Ruihan Yang*, Minghao Zhang*, Nicklas Hansen, Hauzhe Xu, Xiaolong Wang

<https://arxiv.org/abs/2107.03996>

Spotlight

2022

Stabilizing Deep Q-Learning with ConvNets and Vision Transformers under Data Augmentation

Conference on Neural Information Processing Systems (NeurIPS)

Nicklas Hansen, Hao Su, Xiaolong Wang

<https://arxiv.org/abs/2107.00644>

2021

Generalization in Reinforcement Learning by Soft Data Augmentation

International Conference on Robotics and Automation (ICRA)

Nicklas Hansen, Xiaolong Wang

<https://arxiv.org/abs/2011.13389>

2021

Self-Supervised Policy Adaptation during Deployment

International Conference on Learning Representations (ICLR)

Nicklas Hansen, Rishabh Jangir, Yu Sun, Guillem Alenyà, Pieter Abbeel, Alexei A. Efros,

Lerrel Pinto, Xiaolong Wang

<https://arxiv.org/abs/2007.04309>

Spotlight

2021

Short Term Blood Glucose Prediction Based on Continuous Glucose Monitoring Data

IEEE Engineering in Medicine and Biology Conference (EMBC)

Ali Mohebbi, Alexander R. Johansen, Nicklas Hansen, Peter E. Christensen, Jens M. Tarp,

Morten L. Jensen, Henrik Bengtsson, Morten Mørup

<https://arxiv.org/abs/2002.02805>

2020

Teaching

University of California, San Diego

ECE285 Introduction to Visual Learning

· Held office hours, graded assignments.

Teaching Assistant

Spring 2024

Technical University of Denmark

Reinforcement Learning

· Special course that I co-organized w/ Prof. Ole Winther for a group of students. Three weeks of full-time study.

Co-organizer

Jan 2021

Technical University of Denmark

02456 Deep Learning

· Significant course material contributions, supervised 100+ students' projects on reinforcement learning.

02454 Introduction to Cognitive Science

· Assisted tutorial sessions, graded assignments.

Teaching Assistant

Fall 2019, Fall 2020

Fall 2019

Current and Former Mentees

Rishabh Jangir (MS, UCSD -> ML Engineer, Skild AI)

2020 - 2022

Mohit Jain (MS, UCSD -> ML Engineer, Pinterest)	2020 - 2022
Xinyue Chen (BS, NYU Shanghai -> PhD, UC Berkeley)	2021 - 2022
Sateesh Kumar (MS, UCSD -> PhD, UT Austin)	2021 - 2023
Jonathan Zamora-Anaya (BS, UCSD -> MS, USC)	2021 - 2023
Sambaran Ghosal (MS, UCSD -> ML Engineer, BrainCorp)	2021 - 2023
Zirui "Colin" Wang (BS, UCSD -> PhD, UC Berkeley)	2022 - 2023
Ziyan Xiong (BS, Tsinghua University -> PhD, UMass)	2022 - 2023
Yanjie Ze (BS, SJTU -> PhD, Stanford)	2021 - 2023
Yunhai Feng (MS, UCSD -> PhD, Cornell)	2022 - 2024
Chandramouli Rajagopalan (MS, UCSD -> ML Engineer, Streamingo)	2022 - 2024
Jyothir S V (MS, NYU -> RA, NYU)	2023 - 2024
Adrià Lopez (MS, ETH -> PhD, ETH)	2024 - 2025
Nathaniel Thomas (MS, UCSD -> ML Engineer, Assured Robot Intelligence)	2024 - 2026
Yutao Xie (MS, UCSD -> PhD, UC Santa Barbara)	2024 - 2026
Jiajun Xi (MS, UCSD)	2024 - 2026
Varun Giri (MS, Georgia Tech)	2024 -
Anastasiia Pedan (MS, U. of Alberta)	2025 -

Invited Talks

Univ. Michigan	"Robots that learn by doing"	Apr 2026
UC Riverside	"AI agents should learn by doing"	Apr 2026
CMU Intent Lab	"Improving agents via online interaction"	Feb 2026
U. Washington	"Improving agents via online interaction"	Feb 2026
Johns Hopkins U.	"Improving agents via online interaction"	Feb 2026
Stanford REAL	"Improving agents via online interaction"	Feb 2026
NYU CILVR	"Improving agents via online interaction"	Feb 2026
UC Berkeley RLL	"Building Massively Multitask World Models"	Jan 2026
EWM @ NeurIPS	"Massively Multitask World Models for Continuous Control"	Dec 2025
RIWM @ ICCV	"Massively Multitask World Models for Continuous Control"	Oct 2025
Cohere Labs	"World Models on an Academic Budget"	Aug 2025
CU Hong Kong	"Learning World Models for Embodied AI"	May 2025
World Models @ ICLR	"World Models on a Budget"	Apr 2025
UIUC CSLSC	"Learning World Models for Robots"	Feb 2025
KEA Copenhagen	"Data-Driven Algorithms for Robotics"	Dec 2024
USC	"Data-Driven World Models for Robots"	Oct 2024
HuggingFace	"TD-MPC, TD-MPC2, and beyond"	Jun 2024
BeNeRL Seminar	"Data-Driven World Models for Robots"	Jun 2024
TILOS Institute	"Large Datasets and Models for Robots in the Real World"	May 2024
Univ. Michigan	"Robot Learning with (Generalist) World Models"	Jan 2024
Georgia Tech	"Building Generalist World Models"	Jan 2024
Tech. Univ. Denmark	"Data-Driven World Models at Scale: Why, What, and How?"	Dec 2023
Tsinghua IIIS	"The Next Generation of World Models"	Mar 2023
MILA/ServiceNow	"World Models with Behavioral Priors"	Feb 2023
Georgia Tech	"Towards Sample-Efficient Robot Learning with World Models"	Jan 2023
Meta AI (FAIR)	"Pretraining for Control: Current Challenges and Solutions"	Jan 2023
TU Delft	"Model-Based Reinforcement Learning: A Path Towards Generalist Agents?"	Oct 2022
Generally Intelligent	Podcast: https://generallyintelligent.com/podcast/2022-12-16-podcast-episode-25-nicklas-hansen/	Sep 2022
Intel AI	"Temporal Difference Learning for Model Predictive Control"	Apr 2022
Intel AI	"Robots that Generalize"	Aug 2021
G-Research	"Agents that Generalize and Adapt"	Feb 2021

Academic Service

Workshop organizer

2nd Workshop on World Models (**ICLR**) 2026

Journal reviewer

Robotics and Autonomous Systems (**RAS**) 2026

IEEE Transactions on Robotics (**T-RO**) 2024

Transactions on Pattern Analysis and Machine Intelligence (**TPAMI**) 2024

Journal of Machine Learning Research (**JMLR**) 2023

International Journal of Computer Vision (**IJCV**) 2023

Conference reviewer

Conference on Neural Information Processing Systems (**NeurIPS**) 2022, 2023, 2024, 2025, 2026

International Conference on Machine Learning (**ICML**) 2022, 2023, 2025, 2026

International Conference on Learning Representations (**ICLR**) 2024, 2025, 2026

Conference on Computer Vision and Pattern Recognition (**CVPR**) 2022, 2023, 2026

European Conference on Computer Vision (**ECCV**) 2022, 2026

International Conference on Computer Vision (**ICCV**) 2023

IEEE Robotics and Automation Letters (**RA-L**) 2021, 2022, 2023, 2024, 2025

IEEE International Conference on Robotics & Automation (**ICRA**) 2023, 2024, 2026

Annual Conference on Robot Learning (**CoRL**) 2025

International Conference on Intelligent Robots and Systems (**IROS**) 2023, 2025, 2026

Association for the Advancement of Artificial Intelligence (**AAAI**) 2021

Workshop reviewer

Workshop on X-Embodiment Robot Learning @ **CoRL** 2024

Foundation Models for Decision-Making, Workshop @ **NeurIPS** 2023

Self-Supervised Learning - Theory and Practice, Workshop @ **NeurIPS** 2023

Learning Dexterous Manipulation, Workshop @ **RSS** 2023

Structural and Compositional Learning on 3D data, Workshop @ **CVPR** 2023

Self-Supervised Learning - Theory and Practice, Workshop @ **NeurIPS** 2022

Generalizable Policy Learning in the Physical World, Workshop @ **ICLR** 2022

Workshop Presentations

Generalizable Robotic Insertion with World Models

Poster

Workshop on Out-of-Distribution Generalization in Robotics @ **RSS** 2025

Multi-Stage Manipulation with Demonstration-Augmented Reward, Policy, and World Model Learning

Poster

Workshop on Out-of-Distribution Generalization in Robotics @ **RSS** 2025

Musculoskeletal Control with TD-MPC2

Poster

MyoSuite Symposium @ **NeurIPS** (2nd place Locomotion track + Physiological Award) 2024

TD-MPC2: Scalable, Robust World Models for Continuous Control

Poster

Foundation Models for Decision-Making @ **NeurIPS** 2023

Robot Learning @ **NeurIPS** 2023

Pre-Training Robot Learning @ **CoRL** 2023

On Pre-Training for Visuo-Motor Control: Revisiting a Learning-from-Scratch Baseline

Poster

Pre-Training Robot Learning @ **CoRL** 2022

On the Feasibility of Cross-Task Transfer with Model-Based Reinforcement Learning	<i>Poster</i>
Pre-Training Robot Learning @ CoRL	2022
Foundation Models for Decision Making @ NeurIPS	2022
Deep RL Workshop @ NeurIPS	2022
MoDem: Accelerating Visual Model-Based Reinforcement Learning with Demonstrations	<i>Poster</i>
Pre-Training Robot Learning @ CoRL	2022
Deep RL Workshop @ NeurIPS	2022
Look Closer: Bridging Egocentric and Third-Person Views with Transformers for Robotic Manipulation	<i>Poster</i>
Workshop on Deployable Decision Making in Embodied Systems @ NeurIPS	2021
Deep RL Workshop @ NeurIPS	2021
Learning Vision-Guided Quadrupedal Locomotion End-to-End with Cross-Modal Transformers	<i>Poster</i>
Deep RL Workshop @ NeurIPS	2021
Visual Learning and Reasoning for Robotics Workshop @ RSS	2021
Stabilizing Deep Q-Learning with ConvNets and Vision Transformers under Data Augmentation	<i>Poster</i>
Unsupervised RL Workshop @ ICML	2021
Visual Learning and Reasoning for Robotics Workshop @ RSS	2021
Self-Supervised Policy Adaptation During Deployment	<i>Poster</i>
Microsoft Research RL Day	2021
Deep RL Workshop @ NeurIPS	2020
Workshop on Robot Learning @ NeurIPS	2020

Work Experience

NVIDIA Research	<i>Seattle, WA, USA</i>
Research Intern	<i>June 2024 - Feb 2025</i>
· Robot learning. Mentored by Yashraj Narang and Dieter Fox.	
Meta AI (FAIR)	<i>Menlo Park, CA, USA</i>
Research Scientist Intern	<i>June 2022 - Dec 2022</i>
· Model-based reinforcement learning. Mentored by Aravind Rajeswaran.	
raffle.ai	<i>Copenhagen, Denmark</i>
Machine Learning Intern	<i>Summer 2019</i>
· I built and open-sourced a cross-domain text-to-SQL model in PyTorch.	
Retune DSP	<i>Kongens Lyngby, Denmark</i>
Student Assistant	<i>Feb 2019 - Dec 2019</i>
· I helped a team of engineers build and maintain deep learning pipelines for embedded voice control.	
Nordic Transition	<i>Gentofte, Denmark</i>
Student Software Developer	<i>July 2016 - Dec 2019</i>
· I developed and maintained a data management and analysis platform for the HR industry.	

Awards and Scholarships

2026 UC Riverside Future Academics & Mentorship Excellence (FAME) Program	<i>Award</i>
· A 2-day workshop for future faculty applicants (8 recipients nation-wide).	
2024 NVIDIA Graduate Fellowship	<i>Fellowship</i>
· An award of \$60,000 to cover stipend and tuition for 1 year (10 recipients worldwide).	
2023 NeurIPS Top Reviewer	<i>Award</i>
· Awarded free conference registration as a top peer reviewer for the conference.	

2021	Robotics Summer School Scholarship · A scholarship to participate in a two-week summer program in Denmark.	Scholarship
2020	Spar Nord Fond Scholarship · A scholarship to study a semester at UC Berkeley (5 recipients nation-wide).	Scholarship
2020	UC Berkeley's SCET Collider Cup Finalist · Biannual startup competition. Best student project from each class is nominated.	Award
2017	Otto Mønsted Fonds Legat · A grant for students with a GPA ≥ 8.0 who wish to study a semester abroad.	Scholarship

Volunteering

2026	UC San Diego, CSE PhD Visit Days Organizer · Co-organizing visit days for prospective PhD students in the CSE department.	Organizing
2025	NeurIPS, Volunteer · Volunteer work for NeurIPS 2025 in San Diego.	Organizing
2024	Queer in AI, Graduate Application Financial Assistance · Reviewed applications for microgrants associated with graduate applications.	Organizing
2024	UC San Diego, Graduate Women in Computing Mentor · Mentor for first-year PhD students that identify with a gender minority.	Mentorship
2023	UC San Diego, GradAMP Mentor (PhD Applications)	Mentorship
2022	· Supported prospective students through weekly mentor-mentee meetings in Fall.	

Misc. Open-Source Projects

TD-MPC2 Official Implementation (★820) · Public code release for "TD-MPC2: Scalable, Robust World Models for Continuous Control". https://github.com/nicklashansen/tdmpc2	2024
TD-MPC Official Implementation (★513) · Public code release for "Temporal Difference Learning for Model Predictive Control". https://github.com/nicklashansen/tdmpc	2022
DMControl Generalization Benchmark (★189) · Benchmark for generalization in continuous control from pixels. https://github.com/nicklashansen/dmcontrol-generalization-benchmark	2020